

2022
Higher School Certificate
Year 11 Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your student number and/or name at the top of every page

Total marks – 100

Section I – Pages 2–12

20 marks

- Attempt Questions 1–20
- Allow about 30 minutes for this section

Section II – Pages 13–27

80 marks

- Attempt Questions 21–40
- Allow about 1 hour and 30 minutes for this section

This paper MUST NOT be removed from the examination room

STUDENT NUMBER/NAME:

STUDENT NUMBER/NAME:

Section I

20 marks

Attempt Questions 1–20

Allow about 30 minutes for this section

Select the alternative A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
1				
2				
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18				
19				
20				

- 1 The probability that an event occurs is 0.875.

What is the probability that the event will not occur?

- A. 0.125
 - B. 0.225
 - C. 0.045
 - D. 1.25
- 2 Which of these numbers has 3 significant figures?
- A. 2.05×10^3
 - B. 2.050
 - C. 0.025
 - D. 250 000

- 3 If k has the value of -3 , what is the value of $k - k^2$?

- A. -36
- B. -12
- C. 0
- D. 6

- 4 The table provides details of a data set.

Lowest extreme	45
Lower quartile	60
Inter-quartile range	80
Range	125
Median	75

Which of these 5-number summaries could represent this data?

- A. 45 60 75 80 105
- B. 45 60 75 80 170
- C. 45 60 75 140 170
- D. 45 60 75 140 125

STUDENT NUMBER/NAME:

- 5 One week Alex worked 5 hours overtime, paid at time and a half rate. He earned an extra \$285.

If Alex earned a total of \$1235 including overtime, how many hours did he work at the normal rate?

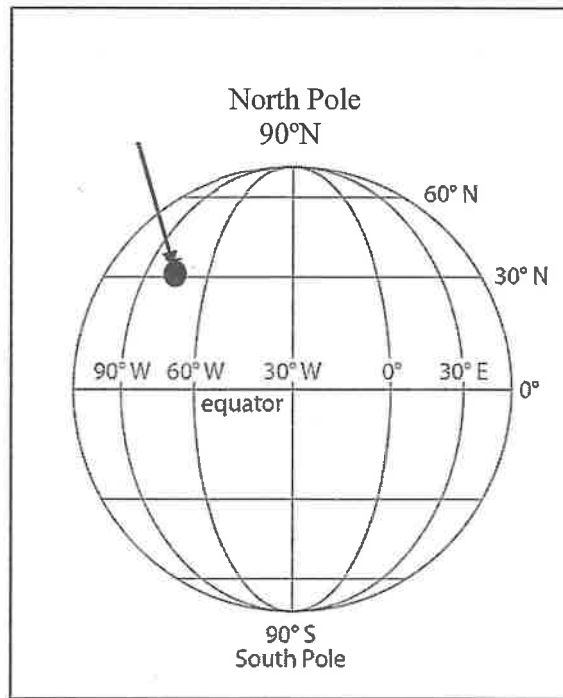
- A. 20
 - B. 25
 - C. 30
 - D. 35
- 6 The likelihood of a disease infecting crops is classified as either very high, high, moderate or low.

What statistical term describes this classification?

- A. Discreet
- B. Ordinal
- C. Continuous
- D. Nominal

STUDENT NUMBER/NAME:

- 7 What is the correct expression of coordinates for the point marked with the arrow on the diagram?



- A. (30°N, 60°W)
B. (65°W, 30°N)
C. (30°N, 75°W)
D. (75°W, 30°N)
- 8 40% of the population in a city have the most common blood type of O⁺.
When two people from the city have blood tests, what is the probability that only one of them will have O⁺ blood?

- A. 10%
B. 20%
C. 24%
D. 48%

STUDENT NUMBER/NAME:

- 9 Jacky's car averages fuel use of 8.5 L/100 km.

Starting with a full tank of 60 litres fuel, Jacky travels 550km.

Approximately how many litres would be left in the tank after this trip?

- A. 9.2
- B. 13.25
- C. 17.9
- D. 46.75

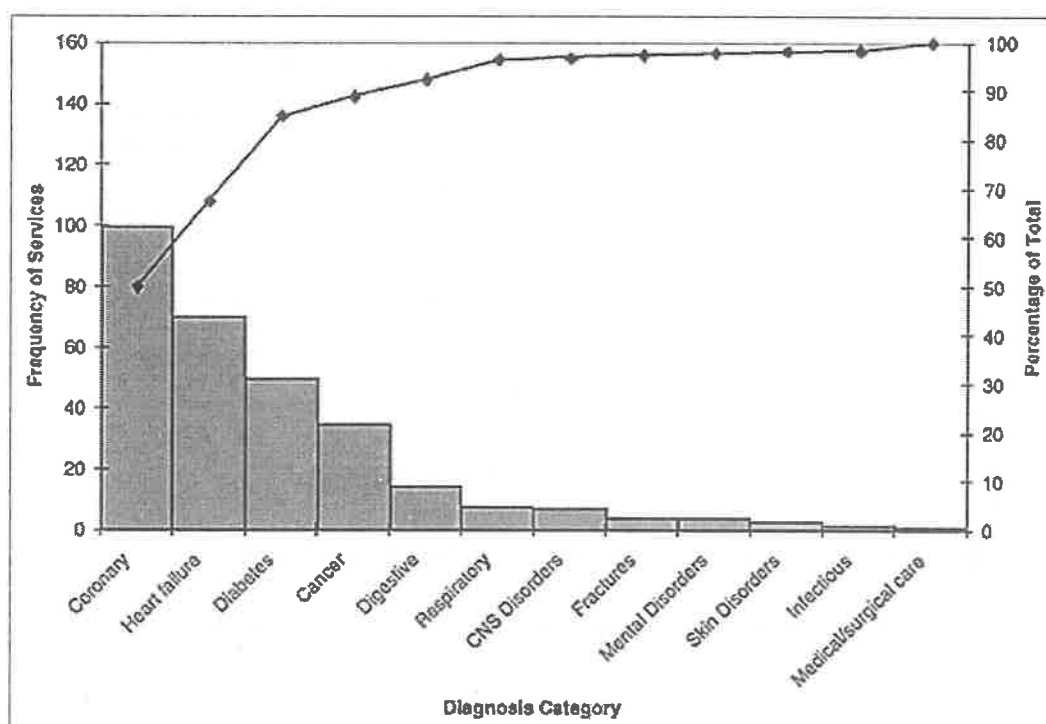
- 10 The mean of a set of 3 numbers is 18 and the mean of another set of 3 numbers is 30.

What is the mean of the 6 numbers?

- A. 2.67
- B. 8
- C. 16
- D. 24

STUDENT NUMBER/NAME:

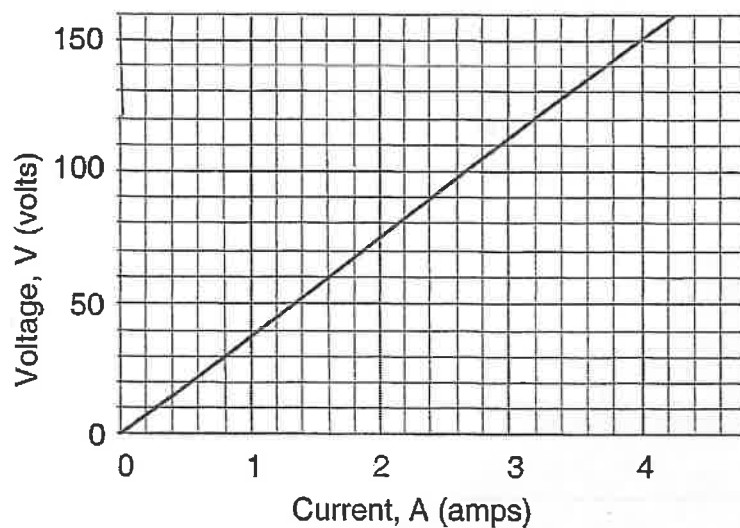
- 11 When patients are admitted to a local hospital for various reasons, data is collected and recorded in the Pareto chart shown below.



According to the chart, what percentage of patients admitted to the hospital during the period of data collection, were diagnosed with diabetes?

- A. 17
- B. 25
- C. 50
- D. 85

- 12 This graph shows the relationship between voltage and current in an electrical circuit.



The voltage in the circuit is decreased from 120 volts to 90 volts.

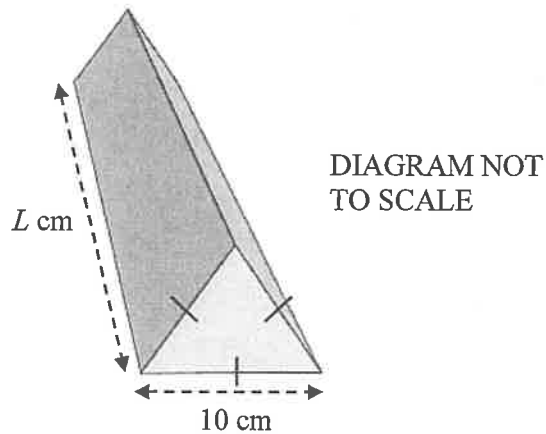
According to the graph, what change in the current occurs as a result?

- A. An increase of 50%
- B. A decrease of 50%
- C. An increase of 25%
- D. A decrease of 25%

13 $\frac{(6y^2)^3}{2y} = ?$

- A. $9y^3$
- B. $27y^3$
- C. $108y^3$
- D. $108y^5$

- 14 The surface area of the solid triangular prism block shown below is approximately 837 cm^2 .



- What is the approximate value of L (the length of the prism)?
- A. 12.5
B. 15
C. 25
D. 26.5
- 15 Four years after a piece of garden equipment was purchased new, it has depreciated in value to \$350 when the straight-line method is applied.
- If the annual rate of depreciation was 15%, what has the equipment lost in value since it was purchased?
- A. \$525
B. \$685
C. \$875
D. \$975

- 16 The formula: $M = \frac{100}{P^2}$ gives the relationship between M and P where P is not zero.

Which of the following correctly gives p as the subject of this formula?

- A. $p = \frac{100}{M^2}$
 B. $p = \frac{10}{M}$
 C. $p = \frac{10}{\sqrt{M}}$
 D. $p = \sqrt{\frac{10}{M}}$

- 17 A piece of a puzzle is in the shape of a semi-circular arch with outer diameter 12 cm and inner diameter 6 cm, as shown in the diagram below.

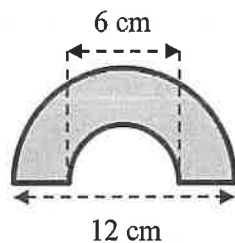
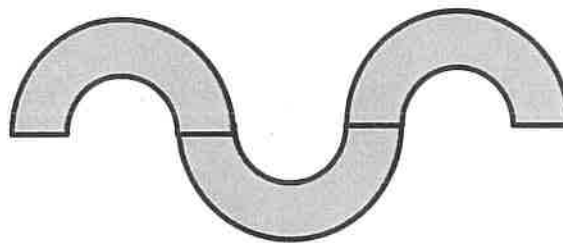


DIAGRAM NOT
TO SCALE

Three of these puzzle pieces are joined together as shown below.



What is the perimeter (cm) of the shape formed from the three pieces?

- A. $6 + 27\pi$
 B. $12 + 27\pi$
 C. $6 + 54\pi$
 D. $12 + 54\pi$

- 18 Samul's vet has recommended a vitamin supplement at the rate of 10 g/L in his pet's drinking water.

Samul purchases the vitamin supplement in solution at his local chemist with the label stating a dosage of "150 g/300 mL".

How many mL of the solution should Samul add to his pet's 500 mL of drinking water?

- A. 10
B. 15
C. 20
D. 90
- 19 The table represents carbon emissions from sectors of an economy and a further breakdown of emissions within the transport sector.

<i>Sector</i>	<i>Percentage</i>	Transport	<i>Percentage</i>
Agriculture	2	Rail	5
Commercial	10	Air	12
Industry	38	Water	6
Mining	5	Trucks/buses	19
Transport	26	Cars	58
Residential	19		

What percentage of the total carbon emissions in the economy are produced from cars?

- A. 14.8
B. 15.1
C. 32
D. 84

STUDENT NUMBER/NAME:

- 20 At a restaurant, a bill was issued to a customer with the items purchased shown.

Entrée:	\$15.00
Main meal:	<u> </u>
Dessert:	\$8.00
G.S.T. (10%):	<u> </u>
Total:	\$61.60

Two of the costs are missing from the bill, the main meal and the G.S.T (which is added to the cost of the three items).

What was the cost of the main meal?

- A. \$31.40
- B. \$32.00
- C. \$32.40
- D. \$33.00

Section II**80 marks****Attempt Questions 21–40****Allow about 1 hour and 30 minutes for this section**

Answer the questions in the spaces provided, using a black pen.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the end of the examination. If you use this space, clearly indicate which question you are answering.

Question 21 (1 mark)**Marks**There is a 7-hour time difference between City P and City Q .**1**The Universal Coordinated Time (UTC) of City Q is +3.If City P is West of City Q , what is the UTC for City P ?

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Question 22 (2 marks)

What interest is earned on an investment of \$10 000 over 4 years at an annual flat rate of 2.2%?

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Question 23 (3 marks)

Jahred records a measurement as 6.25 cm.

(a) What is the lower limit of this measurement?

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(b) What is the percentage error of the recorded measurement?

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Question 24 (2 marks)**Marks**

Maryanne decides to take out a new phone plan which includes a monthly data cost of \$40 and the purchase of a new phone which will cost \$1256, paid in equal monthly instalments. **2**

Maryanne agrees to pay for her plan over 3 years.

If Maryanne obtains a \$320 trade in for her old phone, credited to her account monthly over the 3 years, what amount is paid monthly for her phone plan?

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Question 25 (6 marks)

At two particular bookstores, the change (\$ C) given from \$50 when N books are purchased can be modelled by the equations:

$$C = 50 - 2.5N \text{ (bookstore 1)} \quad \text{and} \quad C = 50 - 4.75N \text{ (bookstore 2)}$$

- (a) What is the change given when 6 books are purchased from bookstore 1? **1**

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- (b) How many books from bookstore 1 were purchased if \$15 change was given? **2**

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- (c) What is the difference in price per book from the two bookstores? **1**

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- (d) What amount of change is given when the maximum number of books are purchased using \$50 from bookstore 2? **2**

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STUDENT NUMBER/NAME:

Question 26 (3 marks)

Marks

The formula for the surface area (S) of a solid hemisphere is given by $S = 3\pi r^2$ where r is the radius.

- (a) What is the surface area of a solid hemisphere with radius 3.25 cm? (Give your answer correct to one decimal place.)

2

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- (b) When two solid hemispheres each, with radius 3.25 cm, are joined to make a solid sphere, what is the surface area of the sphere?

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Question 27 (3 marks)

Solve the equation: $\frac{2k-5}{4} = 3 + \frac{k-2}{3}$.

3

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Question 28 (4 marks)**Marks**

A male person's Blood Alcohol Content (BAC) can be estimated by the formula:

$$BAC = \frac{10N - 7.5H}{6.8M}$$

where N is the number of standard drinks consumed in H hours, and M is the person's weight (kg).

After drinking a number of standard drinks over a period of 4 hours, Ahern has a BAC of 0.06.

- (a) If Ahern weighs 73.5 kg, approximately how many standard drinks did he consume during the 4 hours? 2

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- (b) After a person stops drinking alcohol, the time (T) (in hours) it takes for their BAC to reach zero can be determined by use of the formula: 2

$$T = \frac{BAC}{0.015}$$

Ahern had reached a BAC of zero at 3:00 am on Sunday morning after his drinking the night before.

At approximately what time did Ahern begin drinking alcohol?

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Question 29 (4 marks)**Marks**

Sarina earned an annual gross salary of \$109 850 during the last financial year and was allowed a total of \$4850 in deductions due to her occupation.

4

According to current tax tables, Sarina should have paid \$24 592 in tax during the last financial year based on her annual taxable income and was required to pay an additional 2% of her taxable income as a Medicare levy.

If Sarina paid \$26 000 in tax during the year (P.A.Y.E), is she is entitled to receive a tax refund or does she have to pay further tax? Calculate this amount.

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Question 30 (3 marks)

The diagram shows a circle centre O , of radius 9 cm with a square drawn on top of the circle so that part of the square forms a quadrant of the circle.

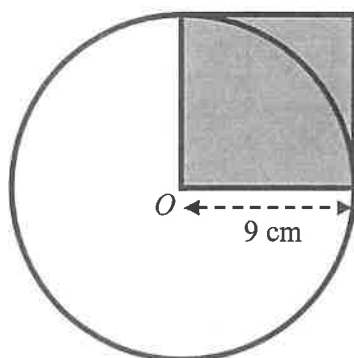
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DIAGRAM NOT
TO SCALE

What percentage of the square's area lies outside the circle?

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Question 31 (6 marks)**Marks**

Two bookstores at different locations in the city are selling a new novel over a period of 10 days.

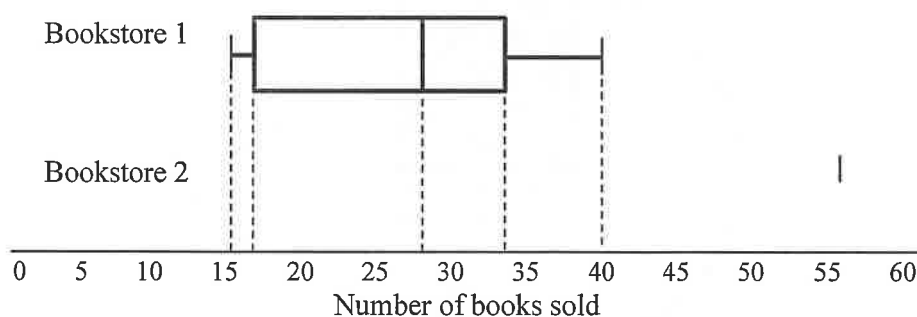
The following back-to-back stem and leaf plot gives the data relating to the number of books sold each day at the two stores.

Bookstore 1		Bookstore 2
9 7 6 6	1	
9 9 7	2	2 2 6 7
4 0	3	1 3 3
0	4	
	5	2 6 6

- (a) Which bookstore sold the highest number of books for a single day? 1

- (b) Which bookstore had the highest median in book sales? 1

- (c) A box and whisker plot for Bookstore 1 is shown below. 3



Construct a box and whisker plot for Bookstore 2 in the space next to its label on the scale above.

- (d) Explain why Bookstore 1 had more consistent sales than Bookstore 2 by referring to the data presented. 1
-

STUDENT NUMBER/NAME:

Question 32 (5 marks)

Marks

When Jennifer was 25 years old, she invested \$30 000 into a compound interest account attracting an annual rate of interest of 3.25% paid half yearly.

When Pierce was 30 years old, he also invested \$30 000 into an account attracting the same rate of compound interest but paid annually.

- (a) When Jennifer and Pierce both reach 65 years old, calculate how much more Jennifer will have in her investment account than Pierce assuming they each only made the one deposit. 4

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- (b) Comment on what these investment strategies might suggest about planning for a financial future. 1

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Question 33 (3 marks)

Ramone earned a commission of \$660 one fortnight after selling merchandise for the company he works for. 3

The commission is based on 12% of sales over \$1500.

What was the total value of the merchandise Ramone sold for his company during this fortnight?

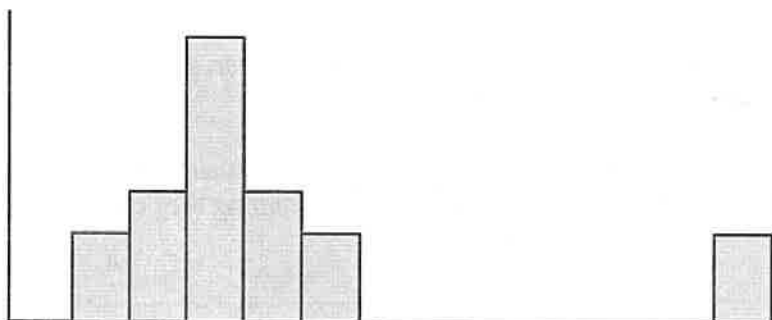
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Question 34 (3 marks)**Marks**

The following data distribution shows the presence of an outlier.



- (a) Describe the distribution if the outlier IS NOT included in the analysis of the data. **1**

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- (b) If the outlier is included in the analysis of the data, how would the distribution be described? **1**

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- (c) If the outlier is included in the analysis of the data, why is the mean not the most appropriate measure to represent the data? **1**

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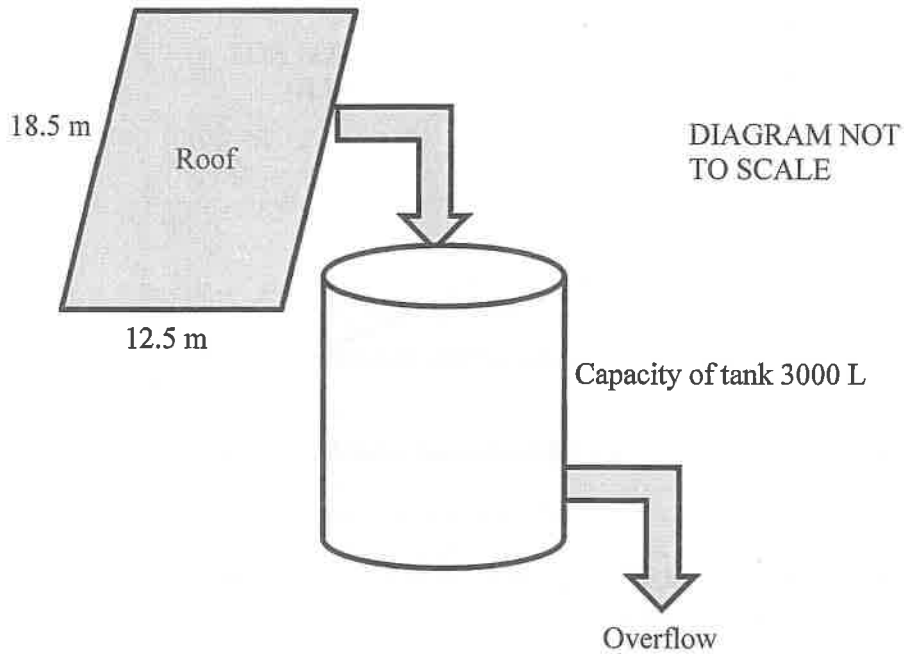
Question 35 (3 marks)**Marks**

15 mm of rainwater falls onto a roof with dimensions $12.5 \text{ m} \times 18.25 \text{ m}$.

3

This rainfall is collected in a 3000-litre capacity water tank, as shown in the diagram.

When the tank is full, a pump engages to push excess water entering the tank out through an overflow pipe.



Assuming the water tank was empty before the rainfall, how much water (if any) overflows from the tank after it is full?

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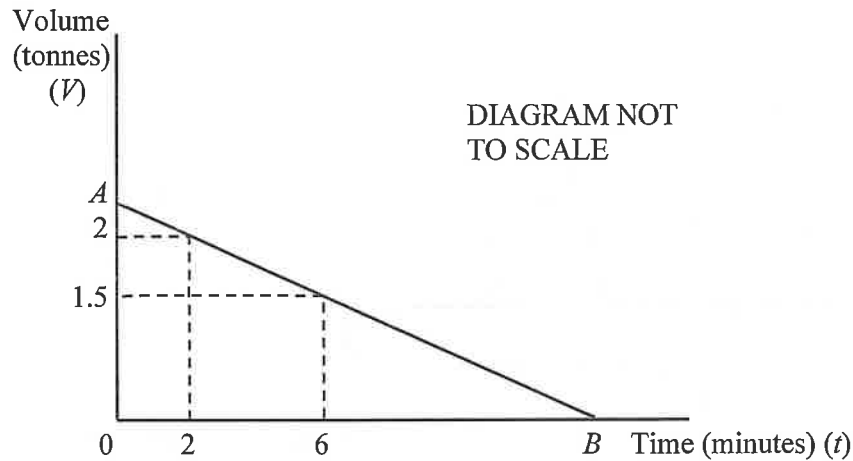
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Question 36 (6 marks)**Marks**

The graph shows the changing amount of wheat in a silo as it is being emptied.

The point A represents the number of tonnes of wheat in the silo when full and the point B represents the time it takes for the silo to empty.



- (a) How many tonnes of wheat are left in the tank after 6 minutes? 1
-
- (b) By using the data presented in the graph, show that the wheat is emptying from the silo at a rate of 125 kg per minute. 1
-
-
-
- (c) By calculating the value of A , write the equation (in terms of V and t), representing the the changing amount of wheat in the silo at any time. 2
-
-
- (d) Determine the value of B on the graph and explain its significance. 2
-
-

Question 37 (5 marks)**Marks**

Students were asked what sport they participate in on weekends during the summer months.

The table below shows the results with the frequency of basketball missing.

<i>Sport</i>	<i>Frequency</i>
Swimming	30
Cricket	60
Tennis	40
Basketball	?
Squash	20

- (a) The relative frequency of basketball is $\frac{2}{5}$.

3

Use the data in the table to determine the relative frequency of tennis.

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- (b) How many students play basketball on weekends?

1

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- (c) A stratified sample of 50 students is to be undertaken to analyse their level of enjoyment in playing their weekend sport.

1

How many students who play cricket on the weekend, should be included in the sample?

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STUDENT NUMBER/NAME:

Question 38 (6 marks)

Marks

In a game, an unbiased dice with the numbers 1-6 is rolled twice.

The product of the two uppermost numbers on the dice represents the score in the game.

- (a) What is the highest score possible in the game? 1

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- (b) What is the probability that the highest score will be thrown? 1

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- (c) What is the probability that a score higher than 20 will be thrown? 2

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- (d) What is the probability that one of the two most likely scores will occur? 2

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Question 39 (7 marks)**Marks**

The diagram shows a trapezium $PQRS$ with parallel sides PQ and SR of lengths l_1 and l_2 , respectively.

The height of the trapezium RT has length h .

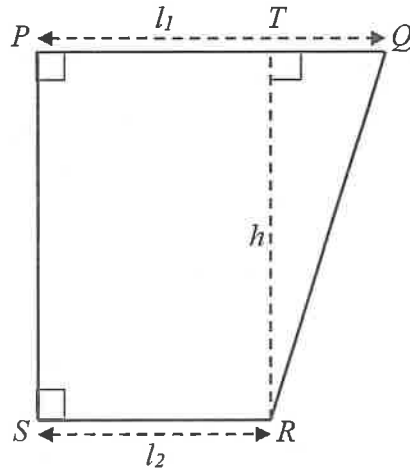


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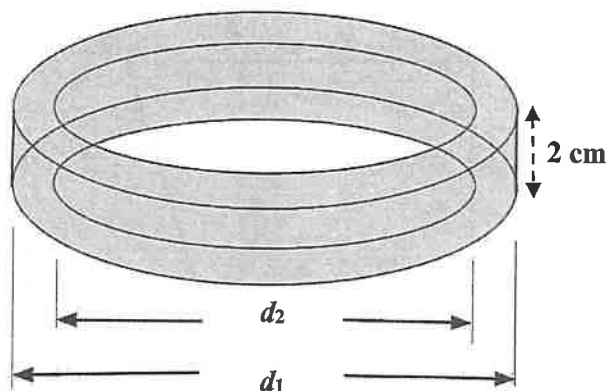
- (a) In terms of l_1 and l_2 , what is the length of TQ ? 1
-
- (b) By considering the area of the rectangle $PTRS$ and triangle TQR , show that the area of the trapezium $PQRS$ can be given by $\frac{h}{2}(l_1 + l_2)$. 3
-
-
-
-
- (c) If the area of the trapezium is 250 cm^2 and $(l_1 + l_2) = 25 \text{ cm}$, what is the value of h ? 2
-
-
-
- (d) If $l_2 = 10 \text{ cm}$, what is the length of TQ ? 1
-

Question 40 (5 marks)**Marks**

A circular rubber seal for a motor has thickness 2 cm.

The outer diameter of the seal is d_1 cm and the inner diameter is d_2 cm, as shown in the diagram below.

DIAGRAM NOT
TO SCALE



- (a) Show that the volume (V) of the seal can be given by: $V = \frac{\pi}{2} (d_1^2 - d_2^2) \text{ cm}^3$.

3

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- (b) The volume of the seal is $18\pi \text{ cm}^3$.

2

If the inner diameter (d_2) = 8 cm, what is the length of the outer diameter d_1 ?

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End of paper

STUDENT NUMBER/NAME:

Section II – Extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

Mathematics Standard 1

Mathematics Standard 2

REFERENCE SHEET

Measurement

Limits of accuracy

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

An outlier is a score

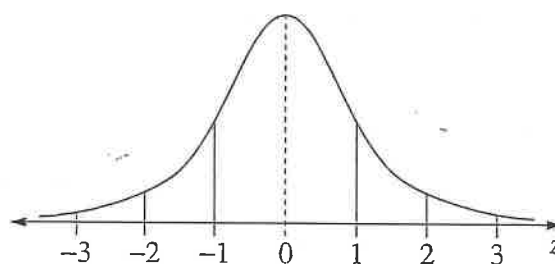
$$\text{less than } Q_1 - 1.5 \times IQR$$

or

$$\text{more than } Q_3 + 1.5 \times IQR$$

$$z = \frac{x - \mu}{\sigma}$$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3